```
'Ke'Shawn Stafford
'Project 1
'02/27/2024

' {$STAMP BS2}
' {$PBASIC 2.5}

SERVO     PIN 15
LCD       PIN 14
FLEX      PIN 13

MIN_TIME  CON  600
MID_TIME  CON  930
MAX_TIME  CON  1445

angle     VAR Byte
time      VAR Word
ledCount  VAR Byte
i         VAR Byte
SV        VAR Word

PAUSE 1000
SEROUT LCD, 84, [22, 12]

DO
    HIGH FLEX
    PAUSE 1
    RCTIME FLEX, 1, time

    IF time < MIN_TIME THEN time = MIN_TIME
    IF time > MAX_TIME THEN time = MAX_TIME

    IF time <= MID_TIME THEN
        angle = (time - MIN_TIME) * 90 / (MID_TIME - MIN_TIME)
    ELSE
        angle = 90 + ((time - MID_TIME) * 90 / (MAX_TIME - MID_TIME))
    ENDIF

    SV = (angle * 5)
    SV = 255 + SV

    PULSOUT SERVO, SV
    PAUSE 20

    ledCount = angle / 18
    IF ledCount > 10 THEN ledCount = 10
    IF angle >= 180 THEN ledCount = 10

    FOR i = 0 TO 9
        IF i < ledCount THEN
            HIGH i
        ELSE
            LOW i
        ENDIF
    NEXT

    SEROUT LCD, 84, [22, 1]
    SEROUT LCD, 84, [128, "Angle: ", DEC angle, " deg"]

    SELECT ledCount
        CASE 1: SEROUT LCD, 84, [209, 218, 220]
        CASE 2: SEROUT LCD, 84, [209, 222]
        CASE 3: SEROUT LCD, 84, [209, 223]
        CASE 4: SEROUT LCD, 84, [209, 225]
        CASE 5: SEROUT LCD, 84, [209, 227]
        CASE 6: SEROUT LCD, 84, [209, 228]
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        CASE 7: SEROUT LCD, 84, [209, 230]
        CASE 8: SEROUT LCD, 84, [209, 219, 220]
        CASE 9: SEROUT LCD, 84, [209, 222]
        CASE 10: SEROUT LCD, 84, [209, 223]
    ENDSELECT

LOOP
```